

Paper 2 - Set 04 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

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Worked solutions

Question 1 CIR-04 · Circles and circle theorems · spec MM3.2

The points $(1, 7)$ and $(5, -1)$ are the ends of a diameter of a circle. Find the equation of the circle.

Answer: D

Centre is the midpoint $(3, 3)$; $\text{radius}^2 = (3 - 1)^2 + (3 - 7)^2 = 4 + 16 = 20$.

Full method - how and why

A diameter's endpoints give the centre (midpoint) and the radius (half the distance) in one move.

1 Centre = midpoint = $\left(\frac{1+5}{2}, \frac{7+(-1)}{2}\right) = (3, 3)$.

why: The centre is the midpoint of any diameter.

2 Radius²: distance from the centre to $(1, 7)$ squared = $(3 - 1)^2 + (3 - 7)^2 = 4 + 16 = 20$.

why: Going from centre to an endpoint is one radius; squaring avoids surds entirely.

3 Write the equation: $(x - 3)^2 + (y - 3)^2 = 20$.

why: The right-hand side is r^2 .

Where it goes wrong

- Using the full diameter length squared (80) as r^2 .
- Writing $r^2 = \sqrt{20}$ - the square root is already undone by squaring.

Takeaway From a diameter: centre is the midpoint, and r^2 is the squared distance from centre to either end.

Question 2 LOG-02 · Exponentials and logarithms · spec MM5.3

Solve $3^{2x} = 4$. Give x in terms of logarithms to base 3.

Answer: B

$$2x = \log_3 4 = 2\log_3 2, \text{ so } x = \log_3 2.$$

Full method - how and why

Solving $a^{kx} = b$. Take logs to the given base and simplify the right-hand side.

1 Take $\log_3$ of both sides: $\log_3 3^{2x} = \log_3 4$, so $2x = \log_3 4$.

why: $\log_3 3^u = u$ - logs and exponentials with the same base undo each other.

2 Rewrite the right-hand side: $\log_3 4 = \log_3 2^2 = 2\log_3 2$.

why: The power law $\log a^k = k\log a$ brings the exponent down.

3 Divide by 2: $x = \log_3 2$.

why: The factor 2 cancels exactly, which is why the answer is so clean.

Where it goes wrong

- Stopping at $x = \frac{1}{2}\log_3 4$ and then mis-simplifying to $\frac{1}{2}\log_3 2$.
- Writing $\log_3 4 = \log_3 2 \times \log_3 2$.

Takeaway $\log a^k = k\log a$; rewriting numbers as powers often makes fractions cancel.

Question 3 POL-15 · Polynomials, factor and remainder theorems · spec MM1.6

The polynomial $x^3 + ax^2 + bx + c$ has roots 2, -1 and 3. Find the value of $a + b + c$.

Answer: D

$p(1) = 1 + a + b + c$, and $p(1) = (1 - 2)(1 + 1)(1 - 3) = (-1)(2)(-2) = 4$. So $a + b + c = 3$. (Directly: $(x - 2)(x + 1)(x - 3) = x^3 - 4x^2 + x + 6$.)

Full method - how and why

You could expand, but $a + b + c$ is the sum of the non-leading coefficients, which $p(1)$ delivers instantly.

1 Note $p(1) = 1 + a + b + c$, so $a + b + c = p(1) - 1$.

why: Substituting $x = 1$ turns every power into 1, so $p(1)$ is the sum of **all** coefficients including the leading 1.

2 Write the factorised form from the roots: $p(x) = (x - 2)(x + 1)(x - 3)$.

why: Monic, because the given polynomial starts with x^3 .

3 Evaluate: $p(1) = (-1)(2)(-2) = 4$.

why: Two negatives and one positive give a positive.

4 So $a + b + c = 4 - 1 = 3$.

why: Remember to subtract the leading coefficient.

Where it goes wrong

- Answering 4 by forgetting the leading 1 in $p(1) = 1 + a + b + c$.
- Expanding and mis-collecting; the expansion is $x^3 - 4x^2 + x + 6$, so $-4 + 1 + 6 = 3$ ✓.

Takeaway $p(1)$ is the sum of all coefficients - a one-second route to questions like this.

Question 4 X2-05 · Straight lines · spec MM3.1

Two distinct lines have equations $y = m_1x + c_1$ and $y = m_2x + c_2$. Which of the following statements is/are true?

I If $m_1m_2 = -1$ then the lines meet at exactly one point.

II If $m_1 = m_2$ then the lines do not meet.

III If the lines meet on the y-axis then $c_1 = c_2$.

Answer: H

I: perpendicular lines are not parallel, so they meet once. II: equal gradients and distinct lines means parallel and different, so no intersection. III: on the y-axis $x = 0$, so $y = c_1 = c_2$.

Full method - how and why

Three claims about two distinct straight lines. Each follows from what gradients and intercepts mean.

1 I: $m_1m_2 = -1$ means the lines are perpendicular, so $m_1 \neq m_2$ and the lines are not parallel - they meet exactly once. **True.**

why: Two non-parallel straight lines always meet at exactly one point.

2 II: $m_1 = m_2$ with the lines **distinct** means $c_1 \neq c_2$: parallel and separate, so they never meet. **True.**

why: The word 'distinct' in the question is doing real work - without it the lines could coincide.

3 III: on the y-axis $x = 0$, so the two y-values are c_1 and c_2 . Meeting there forces $c_1 = c_2$. **True.**

why: Substituting $x = 0$ into both equations.

4 So all three.

why: Match to the list.

Where it goes wrong

- Rejecting II by imagining the lines could coincide - 'distinct' rules that out.
- Rejecting III by thinking they might meet elsewhere; the claim conditions on them meeting **on the y-axis**.

Takeaway Same gradient and distinct $\Rightarrow$ parallel, never meet. Different gradients $\Rightarrow$ exactly one intersection. Read 'distinct' carefully.

Question 5 X2-17 · Integration · spec MM7.5

The trapezium rule with n strips is used to estimate $\int_0^1 \sqrt{x} \, dx$. Which of the following is true for every n ?

Answer: B

$y = \sqrt{x}$ is concave (bends downwards), so every chord lies below the curve and the trapezia underestimate the area.

Full method - how and why

Whether the trapezium rule over- or under-estimates depends on the **concavity** of the curve.

- 1 The trapezium rule replaces the curve on each strip by the straight chord joining the endpoints.
why: Its estimate is the area under those chords.
- 2 $y = \sqrt{x}$ is concave: it rises steeply then flattens, bending downwards.
why: Its second derivative $-\frac{1}{4}x^{-3/2}$ is negative for $x > 0$ - or just picture the shape.
- 3 For a concave curve every chord lies **below** the arc, so each trapezium sits below the true region.
why: That is what concavity means geometrically.
- 4 So the estimate is an underestimate, for every n .
why: The number of strips affects accuracy, not direction.

Where it goes wrong

- Assuming the trapezium rule always overestimates - true only for convex (upward-bending) curves such as $y = x^2$.
- Thinking the parity of n matters - that is Simpson's rule, which needs an even number of strips.
- Answering 'cannot be decided' when the shape settles it.

Takeaway Trapezium rule: convex curve $\Rightarrow$ overestimate, concave curve $\Rightarrow$ underestimate. Sketch and look at the bend.

Question 6 X2-25 · Identifying errors in proofs · spec Err2

A student solves $x^2 - 5x + 6 > 0$ as follows.

I $x^2 - 5x + 6 = (x - 2)(x - 3).$

II So $(x - 2)(x - 3) > 0.$

III So $x - 2 > 0$ and $x - 3 > 0.$

IV So $x > 3.$

Which of the following is correct?

Answer: A

The full solution set is $x < 2$ or $x > 3$.

Full method - how and why

A quadratic inequality solved as if it were an equation-style 'both factors positive'. Find where the missing branch is lost.

1 Lines I and II are fine: $x^2 - 5x + 6 = (x - 2)(x - 3)$, so the inequality is $(x - 2)(x - 3) > 0$.

why: Check: $-2 \times -3 = 6$ and $-2 - 3 = -5$ ✓.

2 Line III says both factors must be positive. But a product is positive when both are positive **or** both are negative. **First error.**

why: Two negatives multiply to a positive - the case the student omits.

3 The missing branch: $x - 2 < 0$ and $x - 3 < 0$, i.e. $x < 2$.

why: Both factors negative requires the smaller bound.

4 Full solution: $x < 2$ or $x > 3$. (Sketch check: the parabola is above the axis outside its roots.)

why: A quick sketch of an upward parabola confirms the two outer intervals.

Where it goes wrong

- Blaming line IV - it correctly follows from line III's (incomplete) conditions.
- Giving the answer as $2 < x < 3$, which is the solution of < 0 , not > 0 .
- Solving factor-by-factor at all; sketching the parabola is faster and safer.

Takeaway $(x - p)(x - q) > 0$ holds **outside** the roots, < 0 **between** them. Sketch rather than case-split.

Question 7 X2-10 · Circles and circle theorems · spec MM3.2

The circle C has centre (a, b) and radius r . P : ' $r > |b|$ '. Q : ' C crosses the x -axis at two distinct points'. Which one of the following is true?

Answer: A

The distance from the centre to the x -axis is $|b|$; the circle cuts the axis twice exactly when this is less than r .

Full method - how and why

Translate the geometry into a distance comparison: circle versus line depends on centre-to-line distance against the radius.

- 1 The perpendicular distance from the centre (a, b) to the x -axis is $|b|$.
why: The x -axis is $y = 0$, so the vertical distance is the size of the y -coordinate.
- 2 If $|b| < r$ the circle reaches across the axis and cuts it twice; if $|b| = r$ it touches once; if $|b| > r$ it misses entirely.
why: Three cases, decided by comparing distance with radius.
- 3 So 'two distinct points' happens exactly when $r > |b|$ - the two statements are equivalent.
why: Necessary **and** sufficient.

Where it goes wrong

- Writing the condition as $r > b$, which fails for a centre below the axis (b negative). The modulus matters.
- Including the tangent case $r = |b|$, which gives one point, not two.

Takeaway Circle versus line: compare the perpendicular distance from the centre with r - less gives two points, equal one, greater none.

Question 8 PRF-17 · Mathematical proof · spec Prf1

Which of the following claims is FALSE, and therefore can be disproved by a counterexample?

Answer: B

$a = 1, b = -2$: $1 > -2$ but $1 < 4$. The others are true.

Full method - how and why

Scan the claims for the one that squaring breaks - negative numbers are where inequalities misbehave.

- 1 ' $a > b \Rightarrow a^2 > b^2$ ' for all real a, b : try $a = 1, b = -2$. Then $1 > -2$ but $1 < 4$. **False**.
why: Squaring reverses order once negatives with larger magnitude are involved.
- 2 ' $a > b > 0 \Rightarrow a^2 > b^2$ ' is **true** - the positivity condition is exactly what the first claim lacked.
why: Comparing the two shows precisely which hypothesis does the work.
- 3 ' $a > b \Rightarrow a + c > b + c$ ' is **true**: adding the same number to both sides preserves order.
why: Valid for every real c , positive or negative.
- 4 ' $a > b \Rightarrow a^3 > b^3$ ' is **true**: cubing is strictly increasing for every real number, unlike squaring.
why: x^3 keeps the sign of x , so no reversal occurs.
- 5 ' $a > b \Rightarrow -a < -b$ ' is **true**: multiplying by -1 reverses the inequality.
why: The standard sign rule.

Where it goes wrong

- Thinking the cubing claim fails - it does not; only **even** powers destroy order.
- Thinking multiplying by -1 keeps the direction.

Takeaway Squaring preserves order only for non-negative numbers; cubing, adding and negating (with a flip) always behave.

Question 9 LGC-17 · The logic of arguments · spec Arg1

Let x be a real number. Which of the following statements is equivalent to ' $x \geq 2$ if and only if $x^3 \geq 8$ '?

Answer: B

'If and only if' is the conjunction of the implication and its converse.

Full method - how and why

Unpacking 'if and only if' into its two implications.

- 1 'A if and only if B' means both 'A if B' ($B \Rightarrow A$) and 'A only if B' ($A \Rightarrow B$).
why: The phrase is literally built from 'if' and 'only if' - each supplies one direction.
- 2 With A: $x \geq 2$ and B: $x^3 \geq 8$, that is $(x \geq 2 \Rightarrow x^3 \geq 8)$ **and** $(x^3 \geq 8 \Rightarrow x \geq 2)$.
why: The conjunction of the implication and its converse.
- 3 (Both happen to be true here, since cubing is increasing - but the question only asks for the equivalent form.)
why: Read what is asked: the structure, not the truth value.

Where it goes wrong

- Giving only one implication - that is half the statement.
- Joining the two with 'or' instead of 'and'.

Takeaway $A \Leftrightarrow B$ is exactly $(A \Rightarrow B)$ **and** $(B \Rightarrow A)$.

Question 10 LGC-24 · The logic of arguments · spec Arg4

Let S be the statement: **every even number greater than 2 can be written as the sum of two primes**. A single counterexample to S would be

Answer: E

To refute 'for all n , $P(n)$ ' you exhibit one n with $P(n)$ false.

Full method - how and why

What it takes to refute a universal statement - and what it takes to refute *this* one in particular.

- 1 The statement has the form 'for all even $n > 2$, $P(n)$ '.
why: Identify the quantifier and the domain: even numbers greater than 2.
- 2 Its negation is 'there exists an even $n > 2$ with $P(n)$ false'.
why: One failure suffices; you do not have to show the property fails often.
- 3 So a counterexample is a single even number greater than 2 that cannot be written as a sum of two primes.
why: It must lie in the stated domain **and** fail the property.
- 4 (This is Goldbach's conjecture - no counterexample is known, which is why it remains a conjecture.)
why: Verifying billions of cases is not a proof; one counterexample would be a disproof.

Where it goes wrong

- Offering an **odd** number that is not a sum of two primes - outside the stated domain, so irrelevant.
- Offering an even number that **is** a sum of two primes - that confirms the statement, it does not refute it.
- Thinking many confirming examples add up to a proof.

Takeaway To disprove 'for all', produce one object **in the domain** for which the property fails; confirming instances never prove it.

Question 11 RSN-20 · Chains of reasoning · spec Prf5

n is a positive integer such that n^2 has exactly 3 positive divisors. Which of the following must be true?

Answer: B

A number with exactly 3 divisors is the square of a prime; so $n^2 = p^2$ and $n = p$ is prime.

Full method - how and why

Use the structure of the divisor count. Exactly three divisors is a very restrictive condition.

- 1 A number with exactly 3 divisors must be p^2 for a prime p : its divisors are $1, p, p^2$.
why: If a number had two distinct prime factors it would have at least 4 divisors; p^3 has 4; p alone has 2.
- 2 So $n^2 = p^2$, and since both are positive, $n = p$.
why: Take the positive square root.
- 3 Therefore n is **prime**.
why: That is the only guaranteed property.
- 4 Check the others: $n = 2$ is even, $n = 3$ is odd - so neither parity is forced; and n prime is never a perfect square.
why: Both $n = 2$ and $n = 3$ satisfy the condition (4 and 9 have three divisors each), which kills the parity options.

Where it goes wrong

- Answering ' n is a perfect square' by confusing n with n^2 .
- Answering $n = 3$ from one example; $n = 2, 5, 7$ all work too.

Takeaway Exactly 3 divisors $\Leftrightarrow$ the square of a prime. Divisor counts come from the prime factorisation exponents.

Question 12 ERR-15 · Identifying errors in proofs · spec Err1

Claim: the sum of the first n odd numbers is n^2 .

- I The first odd number is $1 = 1^2$, so the claim holds for $n = 1$.
- II Suppose the claim holds for some n , i.e. $1 + 3 + \dots + (2n - 1) = n^2$.
- III Adding the next odd number, $2n + 1$, gives $n^2 + 2n + 1 = (n + 1)^2$.
- IV So the claim holds for $n + 1$, and hence for all positive integers n .

Which of the following is correct?

Answer: A

This is a valid proof by induction: a base case, and a step showing that truth for n implies truth for $n + 1$.

Full method - how and why

A correct proof by induction. Check the base case, the inductive hypothesis, the step and the conclusion in turn.

- 1 Line I: base case $n = 1$ gives $1 = 1^2$ ✓.
why: An induction needs a case to start from.
- 2 Line II: assuming the result for some n is the **inductive hypothesis** - not circular, because it is a supposition inside the step, not a global assumption.
why: This is the point most often misidentified as an error.
- 3 Line III: the next odd number after $2n - 1$ is $2n + 1$ ✓, and $n^2 + 2n + 1 = (n + 1)^2$ ✓.
why: Check the term: odd numbers go up in twos, so $(2n - 1) + 2 = 2n + 1$.
- 4 Line IV: base case plus step gives the result for every positive integer ✓. The proof is correct.
why: $n = 1$ gives $n = 2$, which gives $n = 3$, and so on without end.

Where it goes wrong

- Calling line II circular - assuming for **one** n in order to deduce $n + 1$ is exactly how induction works.
- Claiming the next odd number is $2n - 1$; that is the n th one, already included.
- Claiming line IV proves only the $n + 1$ case - the chain from the base case gives all n .

Takeaway Induction = base case + (truth at $n \Rightarrow$ truth at $n + 1$). The inductive hypothesis is not circular reasoning.

Question 13 DIF-11 · Differentiation · spec MM6.3

Which of the following statements about $f(x) = x^3 - 3x$ is/are true?

- I f is increasing for all $x > 1$.
- II f has a local maximum at $x = -1$.
- III $f(x) \geq -2$ for all real x .

Answer: E

$f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$: positive for $x > 1$ (I true); $f'' = 6x < 0$ at $x = -1$ (II true). III false: the local minimum $f(1) = -2$ is not a global minimum since $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$ (e.g. $f(-3) = -18$).

Full method - how and why

Three claims about a cubic. The derivative settles the first two; the third is about global behaviour, where cubics are unbounded.

- 1 Differentiate: $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$.
why: The roots ± 1 are the stationary points.
- 2 I: for $x > 1$ both factors are positive, so $f' > 0$ and f is increasing. **True.**
why: Sign analysis of the factorised derivative.
- 3 II: $f'' = 6x$, so $f''(-1) = -6 < 0$: local maximum at $x = -1$. **True.**
why: Negative second derivative means a local maximum.
- 4 III: the local minimum is $f(1) = -2$, but as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ - for instance $f(-3) = -27 + 9 = -18$. **False.**
why: A **local** minimum is not a global one; odd-degree polynomials are unbounded below.
- 5 I and II only.
why: Match to the options.

Where it goes wrong

- Assuming the local minimum value bounds the whole function.
- Confusing the maximum and minimum positions - for $+x^3$ the maximum is at the smaller x .

Takeaway Cubics have no global maximum or minimum; 'local' claims need f' , 'global' claims need the end behaviour.

Question 14 IND-16 · Indices and surds · spec MM1.2

Which of the following statements is/are true for all positive real numbers a and b ?

I $\sqrt{a}\sqrt{b} = \sqrt{ab}$

II $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$

III $\sqrt{a^2 + b^2} < a + b$

Answer: F

I is a law of surds. II fails for $a = b = 1$: $\sqrt{2} \neq 2$. III: $(a + b)^2 = a^2 + b^2 + 2ab > a^2 + b^2$ since $ab > 0$, and both sides are positive, so III is true.

Full method - how and why

Three surd 'rules', two of which are genuine. The exam skill is knowing which laws distribute and finding a counterexample for the one that does not.

1 Statement I: $\sqrt{a}\sqrt{b} = \sqrt{ab}$ is a law of surds, valid for all $a, b \geq 0$. **True.**

why: Roots distribute over multiplication and division, and the condition $a, b > 0$ in the question is exactly what the law needs.

2 Statement II: test $a = b = 1$: $\sqrt{1+1} = \sqrt{2} \approx 1.41$ but $\sqrt{1} + \sqrt{1} = 2$. **False.**

why: One counterexample is enough to kill a 'for all' statement - and the simplest numbers usually work.

3 Statement III: square both sides. $(a + b)^2 = a^2 + b^2 + 2ab > a^2 + b^2$ since $ab > 0$, and both sides are positive, so $\sqrt{a^2 + b^2} < a + b$. **True.**

why: Squaring is safe here precisely because both sides are known to be positive; that is what makes the comparison valid in both directions.

4 I and III are true: the answer is 'I and III only'.

why: Match your three verdicts to the option list rather than reasoning about the options themselves.

Where it goes wrong

- Assuming roots distribute over addition because they distribute over multiplication - II is the classic error this question is built around.
- Rejecting III because the triangle-inequality picture feels like it should be an equality.

Takeaway Roots split over $\times$ and $\div$ but never over $+$ or $-$; to compare positive quantities, square them.

Question 15 DIF-19 · Differentiation · spec MM6.1

Which of the following statements is/are true for every function f that has a derivative at every real number?

- I If $f'(a) = 0$ then f has a maximum or a minimum at $x = a$.
- II If f has a local minimum at $x = a$ then $f'(a) = 0$.
- III If $f'(x) > 0$ for all x , then $f(3) > f(2)$.

Answer: G

I is false: $f(x) = x^3$ has $f'(0) = 0$ but no turning point. II is true (a stationary point is required at an interior local extremum of a differentiable function). III is true: a strictly increasing function has $f(3) > f(2)$.

Full method - how and why

Three general claims about differentiable functions. The first is the classic false one; the others are standard facts.

- 1 I: test $f(x) = x^3$ at $a = 0$: $f'(0) = 0$ but the function is increasing throughout, so there is no maximum or minimum. **False.**
why: A stationary point can be a point of inflexion.
- 2 II: at an interior local minimum of a differentiable function the derivative must vanish (otherwise you could move downhill in one direction). **True.**
why: This is Fermat's stationary-point theorem, and is why we solve $f' = 0$ in the first place.
- 3 III: if $f' > 0$ everywhere then f is strictly increasing, so $f(3) > f(2)$. **True.**
why: Positive gradient throughout means the value rises as x rises.
- 4 II and III only.
why: Match to the options.

Where it goes wrong

- Accepting I because 'we find maxima and minima by setting $f' = 0$ ' - that finds candidates, not guarantees.
- Doubting II by thinking of endpoints, which the word 'every real number' excludes.

Takeaway $f'(a) = 0$ is necessary but not sufficient for a turning point; inflexions are the counterexample.

Question 16 RSN-15 · Chains of reasoning · spec Prf5

A sequence of 10 integers is such that the sum of any three consecutive terms is positive and the sum of any five consecutive terms is negative. Which of the following statements must be true?

- I The sum of all 10 terms is negative.
- II The sequence contains at least one negative term.
- III The sequence contains at least one positive term.

Answer: H

Terms 1-5 and 6-10 are two blocks of five, each with negative sum (I). A five-block has negative sum, so some term is negative (II). A three-block has positive sum, so some term is positive (III).

Full method - how and why

Group the terms into blocks that the given conditions control, then read off each claim.

- 1 I: split the ten terms into terms 1-5 and terms 6-10 - two blocks of five consecutive terms.
why: Each block has negative sum by hypothesis, and the two blocks together are the whole sequence.
- 2 Negative + negative = negative, so the total is negative. **True.**
why: The grouping is exact - no term is used twice or left out.
- 3 II: terms 1-5 sum to a negative number, so at least one of them is negative (five non-negative terms could not). **True.**
why: A sum is negative only if some summand is.
- 4 III: terms 1-3 sum to a positive number, so at least one is positive. **True.**
why: Same argument the other way round.
- 5 So all three.
why: Match to the list. (Such sequences do exist for 10 terms - the conditions become contradictory only for longer sequences.)

Where it goes wrong

- Trying to construct an explicit sequence; the claims follow from the block structure alone.
- Thinking the sum of all ten is ambiguous because 10 is not a multiple of 3 - use the blocks of five, which do fit.

Takeaway Partition the sequence into blocks the hypotheses control; a negative sum forces a negative term, a positive sum a positive one.

Question 17 ERR-08 · Identifying errors in proofs · spec Err1

Claim: if n is an integer and n^2 is divisible by 4, then n is divisible by 4.

- I Suppose n^2 is divisible by 4.
- II Then $n^2 = 4k$ for some integer k .
- III So $n = 2\sqrt{k}$, and since n is an integer $\sqrt{k}$ is an integer.
- IV Hence n is even, so $n = 2m$, and $n^2 = 4m^2$ is divisible by 4, so n is divisible by 4.

Which of the following is true?

Answer: A

$n = 2$: $n^2 = 4$ is divisible by 4 but n is not. Lines I-III are correct deductions (n even). Line IV concludes ' n divisible by 4' from ' n^2 divisible by 4', which is exactly the unproved claim - circular and false.

Full method - how and why

A circular argument: the last line assumes the very thing being proved. Also, the claim is false.

- 1 Test the claim: $n = 2$ gives $n^2 = 4$, divisible by 4, but 2 is not. **Claim false.**
why: Establish this first; an error must exist.
- 2 Lines I-III: from $n^2 = 4k$, n^2 is even so n is even. The algebra $n = 2\sqrt{k}$ with $\sqrt{k}$ an integer is correct given n is an integer. Valid.
why: These lines legitimately establish that n is even.
- 3 Line IV: it writes $n = 2m$, computes $n^2 = 4m^2$ (true but irrelevant), and then concludes ' n is divisible by 4'. That conclusion does not follow from anything. **First error.**
why: Showing n^2 is divisible by 4 was the **hypothesis**; restating it is not progress. Divisibility of n by 4 would require m even, which is never shown.
- 4 Compare with the working proof for $8 \mid n^2 \Rightarrow 4 \mid n$: there the extra factor forces m even. Here there is no spare factor.
why: Seeing where the analogous proof gets its leverage shows exactly what is missing.

Where it goes wrong

- Accepting line IV because every individual equation in it is true - the **inference** is what fails.
- Blaming line III; introducing $\sqrt{k}$ is unusual but not invalid here.
- Accepting the claim because $4 \mid n \Rightarrow 4 \mid n^2$ (the converse) is true.

Takeaway A step that restates the hypothesis and then asserts the conclusion is circular. $4 \mid n^2$ gives only $2 \mid n$.

Question 18 POL-08 · Polynomials, factor and remainder theorems · spec MM1.6

Which of the following statements about the polynomial $p(x) = x^3 - 6x^2 + 11x - 6$ is/are true?

- I $p(x)$ has three distinct real roots.
- II $(x - 4)$ is a factor of $p(x) - 6$.
- III $p(x) > 0$ for all $x > 3$.

Answer: H

$p(x) = (x - 1)(x - 2)(x - 3)$, so I is true. $p(4) - 6 = 3 \cdot 2 \cdot 1 - 6 = 0$, so II is true. For $x > 3$ all three factors are positive, so III is true.

Full method - how and why

Three statements about one cubic. Factorising it first makes all three immediate.

- 1 Factorise: $p(1) = 1 - 6 + 11 - 6 = 0$, so $(x - 1)$ is a factor and $p(x) = (x - 1)(x - 2)(x - 3)$.
why: Test the divisors of 6; the resulting quadratic $x^2 - 5x + 6$ factorises easily.
- 2 I: the roots 1, 2, 3 are distinct. **True.**
why: Three distinct linear factors means three distinct roots.
- 3 II: $(x - 4)$ is a factor of $p(x) - 6$ iff $p(4) - 6 = 0$. Now $p(4) = 3 \times 2 \times 1 = 6$, so $p(4) - 6 = 0$. **True.**
why: Factor Theorem applied to the shifted polynomial - evaluate the whole expression $p(x) - 6$ at $x = 4$.
- 4 III: for $x > 3$ each factor $x - 1$, $x - 2$, $x - 3$ is positive, so the product is positive. **True.**
why: Sign analysis from the factorised form; beyond the largest root everything is positive.
- 5 All three: 'I, II and III'.
why: Match verdicts to options.

Where it goes wrong

- Expanding $p(x) - 6$ and re-testing from scratch instead of just evaluating $p(4) - 6$.
- Checking III at one value like $x = 10$ and calling it proved - the factor argument covers every $x > 3$.

Takeaway Factorise first: roots, sign and shifted-factor questions then all read off the same form.

Question 19 LGC-12 · The logic of arguments · spec Arg3

Consider the statement: **there exists a real number k such that for all real x , $x^2 + kx + 1 > 0$** . Which of the following is true?

Answer: A

Need discriminant $k^2 - 4 < 0$, i.e. $-2 < k < 2$. So some (not all) values of k work; $k = 0$ gives $x^2 + 1 > 0$. ($k = 2$ gives $(x + 1)^2$, which is 0 at $x = -1$.)

Full method - how and why

An $\exists\forall$ statement: you only need to produce **one** k that works for **all** x . Then check why the offered reasons are right or wrong.

1 Try $k = 0$: $x^2 + 1 > 0$ for every real x , since $x^2 \geq 0$.

why: One successful witness proves an existential statement.

2 So the statement is true. Now check the reason offered with each claim.

why: In a 'true because...' question both the verdict **and** the justification must be right.

3 Is it true for **every** k ? No: the condition is discriminant < 0 , i.e. $k^2 - 4 < 0$, so only $-2 < k < 2$.

why: A positive quadratic with no real roots never crosses the axis.

4 And $k = 2$ gives $(x + 1)^2$, which equals 0 at $x = -1$ - so that witness fails the strict inequality.

why: The boundary case of the discriminant gives ≥ 0 , not > 0 .

5 So: true, with $k = 0$ as the witness.

why: Match to the list.

Where it goes wrong

- Claiming it holds for all k - $k = 5$ gives $x^2 + 5x + 1$, which is negative at $x = -1$.
- Offering $k = 2$ as the witness, which fails only at the single point $x = -1$ - easy to miss.
- Thinking a quadratic with a positive x^2 coefficient must eventually be negative; it is large **positive** for large $|x|$.

Takeaway To prove $\exists$, exhibit one witness - but check the witness satisfies the condition **strictly** where required.

Question 20 RSN-02 · Chains of reasoning · spec Prf5

The integers a, b, c satisfy $a + b + c = 0$ and $abc = 12$. Which of the following statements must be true?

- I Exactly one of a, b, c is positive.
- II $a^2 + b^2 + c^2$ is even.
- III One of a, b, c is divisible by 3.

Answer: H

$abc > 0$ means either none or exactly two of them are negative; they cannot all be positive with sum 0, so exactly two are negative and one positive (I). $a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + bc + ca) = -2(ab + bc + ca)$ is even (II). 3 is prime and divides abc , so it divides one of them (III). (Example: $-1, -3, 4$.)

Full method - how and why

Three claims about integers constrained by a sum and a product. Sign analysis, an algebraic identity, and a prime-divisibility fact.

- 1 I: $abc = 12 > 0$, so the number of negatives is even: 0 or 2. If none were negative, three positives could not sum to 0. So exactly two are negative and one positive. **True.**
why: A positive product means an even count of negative factors; note 0 is impossible since $abc \neq 0$.
- 2 II: use $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$. With $a + b + c = 0$, $a^2 + b^2 + c^2 = -2(ab + bc + ca)$.
why: The identity converts a sum of squares into twice an integer.
- 3 That is $2 \times$ an integer, hence **even. True.**
why: $ab + bc + ca$ is an integer because a, b, c are.
- 4 III: 3 is prime and $3 \mid 12 = abc$, so 3 divides one of the factors. **True.**
why: Euclid's lemma: a prime dividing a product divides one of the factors. This would fail for a composite like 4.
- 5 So all three. (Check with $-1, -3, 4$: sum 0 ✓, product 12 ✓.)
why: A concrete example confirms the constraints are satisfiable at all.

Where it goes wrong

- Assuming exactly one negative because the product is positive - that would make the product negative.
- Trying to settle II by testing examples rather than using the identity.
- Thinking III needs a specific example; it follows from primality alone.

Takeaway $a + b + c = 0$ makes $a^2 + b^2 + c^2 = -2(ab + bc + ca)$; and a prime dividing a product must divide a factor.